Assignment 1: Group 4

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TASK 1: Summarize Complete Journeys

A "Success" is defined as any journey containing at least one **order_shipped** event (Event ID: 28), and an unsuccessful for one more than 60 days with inactivity. Success represents **19.53%** (279,363 users) of the total population.

The following results highlight the difference in actions and days between users who convert and those who do not.

```
Top actions for Successful Users...
shape: (10, 3)
```

event_name	len	stage
---	---	---
str	u32	str
browse_products	6334496	First Purchase
view_cart	2285225	First Purchase
add_to_cart	1347195	First Purchase
application_web_view	1093096	Apply for Credit
begin_checkout	968987	First Purchase
place_downpayment	344880	Downpayment
order_shipped	279363	Order Shipped
account_activitation	279298	Credit Account
account_downpaymentcleared	277635	Downpayment
application_web_approved	269878	Apply for Credit

```
Total Unique Users: 1,430,445
Successful Users (Orders): 279,363
Conversion Rate: 19.53%

--- Success vs. Non-Success Comparison ---
shape: (2, 5)
```

is_success	median_actions	avg_actions	median_days	avg_days
---	---	---	---	---
bool	f64	f64	f64	f64
false	21.0	31.782901	159.337031	154.369367
true	41.0	54.639075	17.96625	48.283446

From this last plot, we hypothesize that successful cases have "high momentum", we define this as the amount of action in the first day of the journey.

is_success --- bool	avg_early_momentum --- f64
true	17.417367
false	11.121438

TASK 2:

In this dataset, "Incomplete" (Unsuccessful) journeys are defined by a lack of action for 60 days, whereas "Successful" journeys end with an order_shipped event. However, a "completed" successful journey is often much shorter in duration than the 60-day threshold required to declare a failure.

To create a fair comparison and avoid a look-ahead Bias (predicting the past using future information), we implemented two specific strategies: **Proportional Weighting and Random Truncation**.

Successful journeys in our data have a median duration of 18 days, while unsuccessful journeys linger for a median of 159 days.

In a real-world environment, we are much more likely to "catch" a long journey in progress than a short one. Logically, a user in a 100 day journey is 100 times more likely to be int he testing set than a use rin a single day journey.

To ensure our training data reflects this "Testing Distribution," we did not sample users uniformly. Instead we create a feature called sampling_weight, which will create "importance" for longer journeys in our machine learning model.

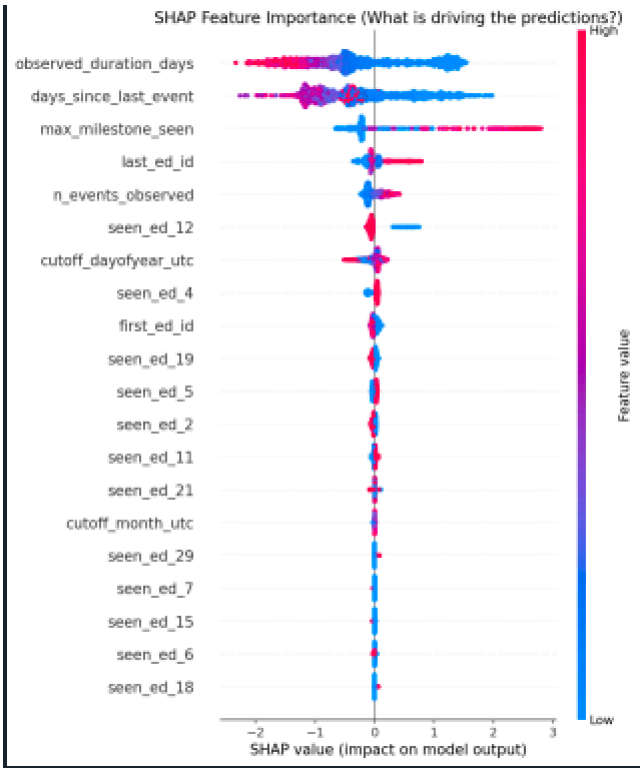
Finally, we apply the Random Truncation, to simulate the random ability to be caught in the middle of a journey, rather than including the successful actions itself.

Task 3:

```
================================================================================
                        FINAL MODEL COMPARISON (Validation Set)
================================================================================
Model                  | Brier (↓)  | Log Loss (↓) | ROC-AUC (↑) | PR-AUC (↑) | Accuracy
--------------------------------------------------------------------------------
Logistic Regression    | 0.0842     | 0.2942       | 0.8971      | 0.7763     | 0.8826
XGBoost (Calibrated)   | 0.0780     | 0.2676       | 0.9136      | 0.8081     | 0.8965
Neural Network         | 0.0780     | 0.2627       | 0.9109      | 0.8020     | 0.8953
================================================================================
```

In this, we compare the results of three separate models. We used a basic logistic regression, XGBoost, and a neural network. We can see that the XGBoost and neural network performed very similarly, while

the logistic regression performed slightly worse.



For the feature importance plot, we opted to use SHAP and the XGBoost model for our analysis. Our top three influential features were observed_duration_days, days_since_last_event, and max_milestone_seen. Not a single count for each event definition made the top 20, so we can likely conclude that they are not an essential feature.